

EXPERIMENT 7

7.1.sci (C:\Users\naikn\Documents\7.1.sci) - SciNotes

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```
1 function y = sinc_interp(x, t, Ts)
2 y = 0*t;
3 for k = 1:length(x)
4 y = y + x(k) * sinc((t - (k-1 - length(x)/2)*Ts) / Ts);
5 end
6 endfunction
7 t = -0.02:1e-4:0.02;
8 f0 = 300;
9 x = sin(2*pi*f0*t);
10
11 fs = 2000; Ts = 1/fs;
12 n = -40:40;
13 ns = n*Ts;
14 xs = sin(2*pi*f0*ns);
15 x_rec = sinc_interp(xs, t, Ts);
16 clf;
17 plot(t, x, 'k');
18 plot(t, x_rec, 'r--');
19 legend("Original", "Reconstructed");
20 xtitle("Sinc-Reconstruction");
21
```



